

Understanding ADC Bits and Effective Number of Bits (ENOB)

Oscilloscope signal integrity: Part 1

ADC Bits vs. Effective Number of Bits

The number of analog-to-digital converter (ADC) bits in an oscilloscope is one of the most widely known specifications. Consequently, many engineers rely on this as the sole specification that determines an oscilloscope's quality. However, the importance of ADC bits is often exaggerated while other critical indications of signal integrity get pushed to the background.

Just as important as the number of ADC bits is the system's effective number of bits (system ENOB). System ENOB is the number of bits that are actually effective while making a measurement. In any oscilloscope, some number of ADC bits will be useless — they only operate in the noise. Therefore ENOB, not ADC bits, determines the quality of measurements you are able to make on that oscilloscope. If the measurement quality is too poor, the results are inaccurate and non-repeatable. This can lead to incorrect assumptions in your design.

ENOB is a better indication of signal integrity because it takes system error into account.

Many engineers have not heard of system ENOB. This specification isn't often mentioned simply because designing for a high ENOB is not as easy as putting in a high-bit ADC. The front end and supporting circuitry around the ADC must be designed with a high degree of quality, which is no simple task.

Vendors naturally market the specifications that make them look best. So a high ADC specification is important. However, you need to make sure you are looking at the other important components that go into signal integrity as well. The ADC is just one part of the overall system.

Why You Should Care About ADC Bits and ENOB

The number of ADC bits is important because of quantization levels, or Q levels. The more bits there are, the more Q levels there can be. The more Q levels there are, the higher the accuracy because the oscilloscope can capture and display more of the true signal detail.

When an ADC converts a signal from analog to digital, it can convert only at certain levels. Using Figure 1 as an example, think about a 2-bit ADC. You have $2^2 = 4$ levels to quantize the signal: 11, 10, 01, and 00.

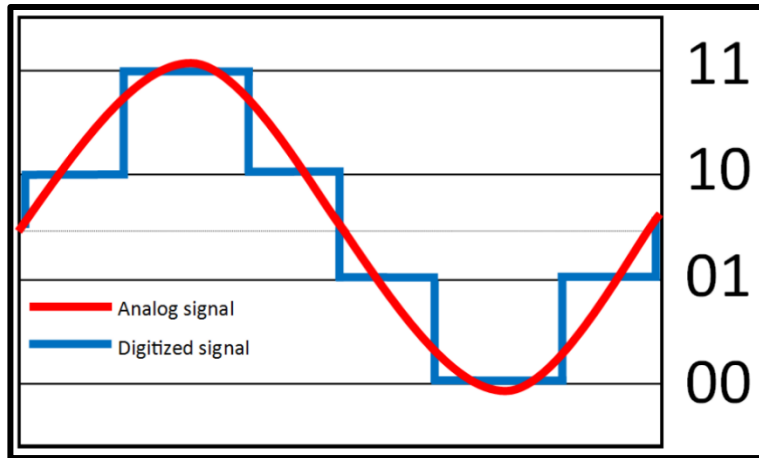


Figure 1. Visualization of quantization levels for a 2-bit ADC.

Now, compare that to a 3-bit ADC. This gives the oscilloscope 23, or 8, levels to work with. If you are measuring the same signal with the 2-bit ADC and 3-bit ADC, you can see in Figure 2 how much more detail the 3-bit ADC provides. The oscilloscope can convert more of your original analog signal. That results in significantly more signal detail when it is digitally reconstructed and shown on the screen.

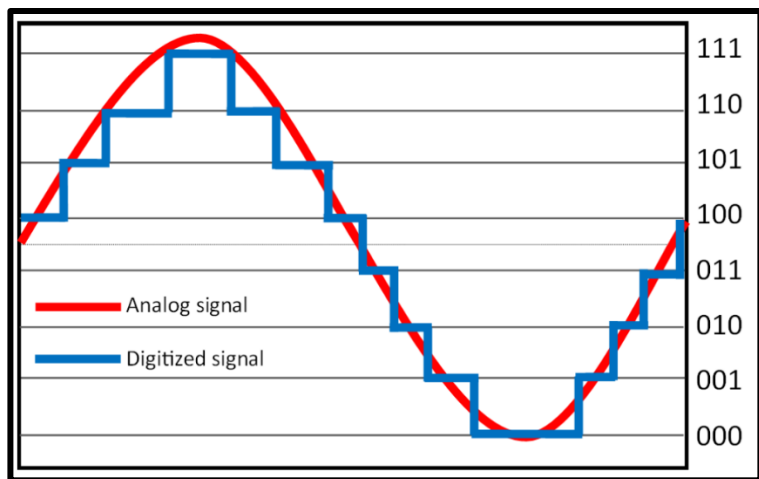


Figure 2. Visualization of quantization levels for a 3-bit ADC.

With high-performance oscilloscopes like the Keysight Infiniium S-Series, the 10-bit ADC should allow you to achieve high accuracy with 1024 Q levels. However, remember that accuracy also depends on the design of the rest of the oscilloscope.

The rest of the measurement system can make the effective number of bits (ENOB) very different from the ADC bit specification. The S-Series uses a 10-bit ADC. Figure 3 shows the ENOB data, which averages around 8 bits. How can the ENOB only be 8 bits when we have a 10-bit ADC?

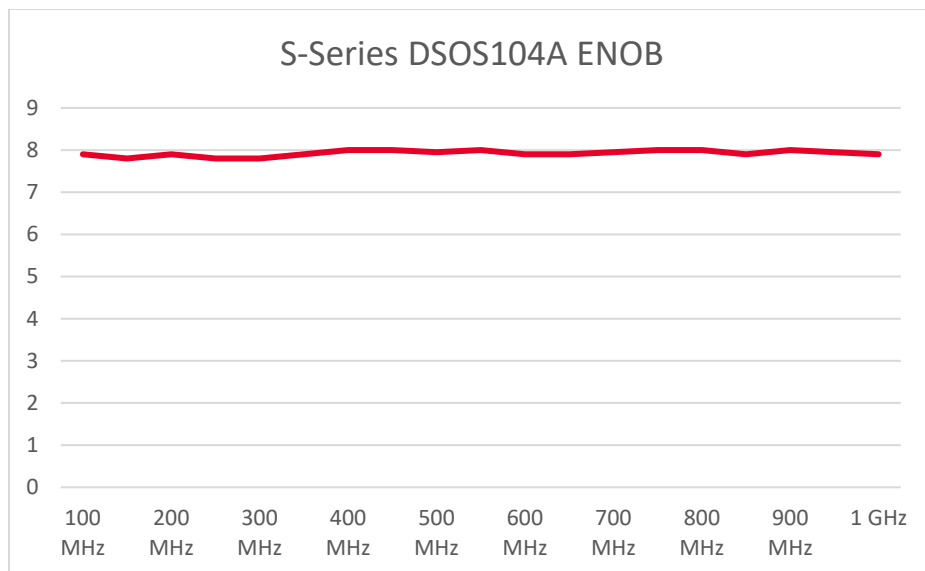


Figure 3. The ENOB of the **S-Series** DSOS104A 1 GHz real-time oscilloscope from 100 MHz to 1 GHz hovers around an average of 8 bits, ensuring you always achieve the highest signal integrity.

Take a 1.8 Vpp signal as an example. To measure that signal, the vertical scale should be set to about 2 V full screen. With a 10-bit ADC, theoretically you get a sensitivity of 1.95 mV per quantization level. In other words, you should get 1.95 mV of vertical resolution.

However, because of the other components in the measurement system, not all 10 of those ADC bits will actually be useful in your measurements. Noise and distortion primarily impact the number of effective bits. If the oscilloscope's random noise is larger than each quantization level — in this case, if noise is greater than 1.95 mV — the scope cannot differentiate that noise from the actual signal. This basically makes that Q-level irrelevant. This, in turn, decreases the number of useful Q-levels, and therefore reduces the quantity of bits that are effective in making your measurement.

There is no way to build an oscilloscope that does not have some noise. It's impossible. There will always be noise, and the noise will always affect your number of bits. That is why we look at effective number of bits rather than just ADC bits. Taking both specifications into account will give you a better idea of the instrument's overall measurement quality.

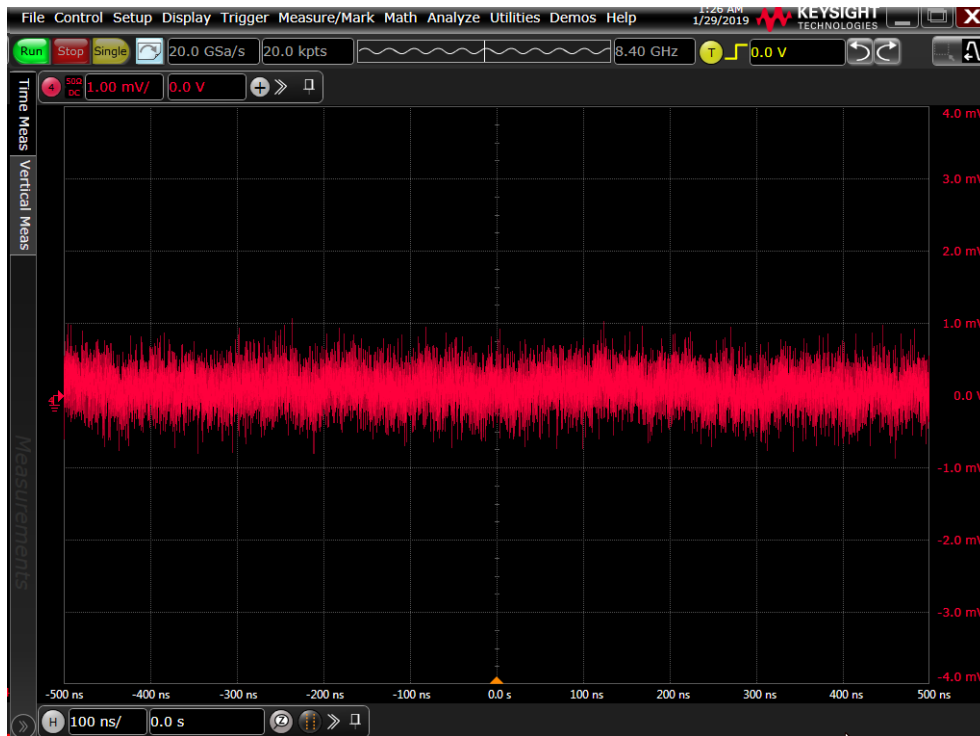


Figure 4. The S-Series experiences only 90 μ V of noise at 1 mV/div and 1 GHz bandwidth.

This means it is also very important to analyze the baseline noise of an oscilloscope to be sure it is low. However, note that factors other than noise influence ENOB. Although noise is the largest contributor to a lower ENOB specification, other distortions in the system can also affect it.

Another thing to keep in mind is that the ENOB specification does not consider errors from things such as the frequency response, gain accuracy, or offset accuracy. Therefore, it is important to analyze those contributors separately to get a complete picture of the oscilloscope's accuracy.

System ENOB vs. ADC ENOB

Another important nuance is the difference between system ENOB and ADC ENOB. Since system ENOB is typically not very flattering, some vendors will list ADC ENOB on their data sheets. Some people see the term “ENOB” and take that at face value. However, there is more to the story.

ADC ENOB refers to the effective number of bits in the ADC, and only the ADC. However, we already learned an oscilloscope comprises much more than an ADC. There are other components that are equally important in the measurement system, including the probing solution.

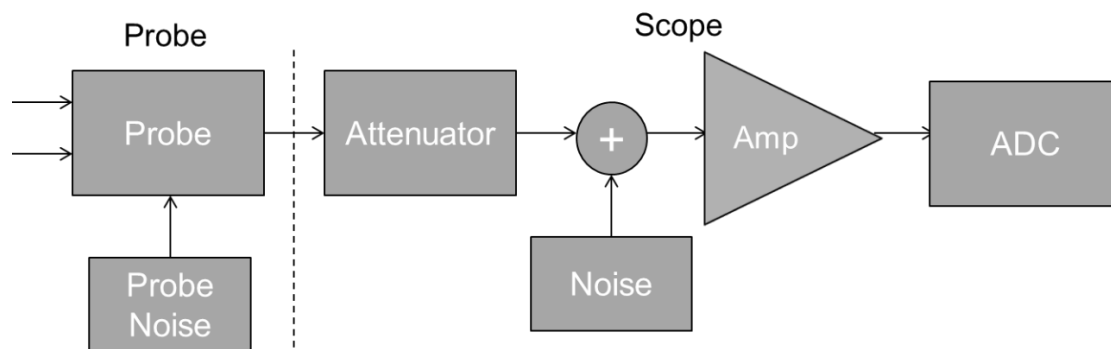


Figure 5. The signal path of an oscilloscope from the probe to the ADC.

ADC ENOB does not represent the number of bits the entire oscilloscope measurement system is using, and that is what matters when you make measurements. The system ENOB specification is the number of bits that convey information when displaying and measuring a signal.

If you see system ENOB listed as a single number, note that this is an average. When you ask for this data, be sure you receive a plotted series of data based on various frequencies within the oscilloscope's bandwidth range, as shown in Figure 3. This way you can ensure that the oscilloscope has sufficient ENOB across the entire bandwidth spectrum.

It is also possible to measure the system ENOB of your oscilloscope with a PSG and a MATLAB script. However, it is best to work with a Keysight representative to tailor the script to your specific oscilloscope model and bandwidth.

Summary

The moral: do not base your buying decision on the ADC specification alone. The number of ADC bits is important because it means more quantization levels. However, that is irrelevant if the rest of the oscilloscope is not designed properly.

Understanding ENOB and what it means to your measurements is critical when creating designs that have the accuracy you and your customers need. High ENOB provides more accurate waveform shapes, more accurate and repeatable measurements, wider eye diagrams and less jitter.

In general, by choosing an oscilloscope with superior ENOB, you are choosing a scope with better signal integrity. However, make sure you are also considering other important specifications such as baseline noise and offset accuracy.

Learn more about these important considerations in determining signal integrity by downloading the eBook, [How to Determine Oscilloscope Signal Integrity](#).

For Part 2 of the Oscilloscope Signal Integrity series, read the white paper, [Everything You Need to Know About Sample Rate](#).



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